LUIS E. ESCAREÑO FERNÁNDEZ

MECHATRONICS ENGINEER

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About Me

Mechatronics Engineering graduate with a strong interest in intelligent robotics, mechanical design, and applied electronics. Passionate about developing technological solutions that integrate automation, artificial intelligence, and embedded systems to solve real-world problems. Seeking opportunities to grow professionally, contributing innovative ideas and acquiring new skills in the fields of robotics, electronics, and applied engineering.

Skills

- Mechanical Design
- Manufacturing Processes
- Problem Solving
- Teamwork
- Python and C Programming

Professional Experience

Freelance Mechatronics Engineer *2024 – Present*

Fiverr

- Designed and developed over 20 custom PCBs for international clients, focusing on IoT applications.
- Programmed embedded systems (ESP32, STM32) for automated robotics projects.
- Managed client projects from initial concept and simulation to final prototyping and delivery.

Education

B.S. in Mechatronics Engineering *2020 – 2025*

National Polytechnic Institute - UPIIZ Campus (Mexico)

Projects

Mobile Agricultural Robot for Damage Detection in Bean Crops *2024 – 2025*

Undergraduate Thesis

Developed a mobile robot using computer vision algorithms to autonomously navigate through bean crop rows and detect damage caused by pests or nutrient deficiencies.

Fire-Fighting Robot *2023 – 2024*

Academic Project

Designed and built a scale robot capable of remote-controlled navigation to detect and extinguish small fires in residential settings.

Certifications

PROFESSIONAL | SOLIDWORKS *2023*

- Mechanical Design (CSWP)
- Weldments (CSWPA-WD)
- Surfacing (CSWPA-SU)
- Sheet Metal (CSWPA-SM)

ASSOCIATE | SOLIDWORKS *2023*

- Mechanical Design (CSWA)
- Additive Manufacturing (CSWA-AM)
- Electrical Design (CSWA-E)
- Simulation (CSWA-S)

Courses

MATLAB - MATHWORKS *2022*

- Deep Learning Onramp
- MATLAB Onramp
- Simulink Onramp

Intelligent Systems *2024*
UPIIZ-IPN

Introduction to ROS and Autonomous Driving with Donkietown *2023*
Center for Research in Mathematics (CIMAT)

Webinar: Introduction to Bionic Exoskeletons *2022*
P4H Bionics

Publications

"Design Validation of an Agricultural Mobile Robot for the Detection of Damaged Areas in Bean Crops" *Oct. 2024*
23rd National Congress of Mechatronics (Querétaro, Mexico)

"Deployment of an Agricultural Mobile Robot for the Detection of Damaged Areas in Bean Crops Using Artificial Intelligence" *July 2025*
5th International Conference on Electrical, Computer and Energy Technologies (Paris, France)

Conferences

"Design Validation of an Agricultural Mobile Robot for the Detection of Damaged Areas in Bean Crops" *Oct. 2025*
Speaker at the 23rd National Congress of Mechatronics (Querétaro, Mexico)

"Service Robotics: Intelligent Solutions for a Better World" *Jan. 2024*
Keynote Speaker at the 20th Anniversary of Zigzag (Zacatecas, Mexico)

Conference Presentation *July 2025*
5th International Conference on Electrical, Computer and Energy Technologies (Paris, France)

"Deployment of an Agricultural Mobile Robot for the Detection of Damaged Areas in Bean Crops Using Artificial Intelligence"

Languages

- **Spanish:** Native
- **English:** Advanced